SkillSync

Workforce capacity and skill gap intelligence · Demo company: TCS-style delivery unit · 60 employees, 5 projects, 12 weeks (synthetic)

Headcount	Peak demand	Unfilled FTE-weeks	Projects at risk
60	56 FTE (93%)	79	3 of 5

Plan cost
₹13.5 L

1 · Weekly unfilled skill demand (FTE)

View: Current workforce

	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
Cloud						2	2	2	2	2		
Data Eng			1	3	3	3	3	3	3	3	3	3
ML/AI				4	4	4	4	4	4	4	4	4
Cybersecurity						1	1	1	1	1		
DevOps												
Java												
React												
QA Automation												

Red cells = project work that has no qualified, free person that week. Blank = fully staffed.

2 · Skill supply vs demand

Skill	Qualified staff	Peak demand	Peak gap	Gap FTE-weeks	Avg qualified but idle
Cloud	11	10	2	10	2.2
Data Eng	8	11	3	28	1.3
ML/AI	4	8	4	36	0.8
Cybersecurity	7	8	1	5	1.7
DevOps	10	6	–	–	3.8
Java	16	5	–	–	9.1
React	9	5	–	–	5.4
QA Automation	13	3	–	–	10.4

3 · Recommended actions: internal allocation, training, hiring

Skill gap	Peak gap (FTE)	First gap	Weeks short	Train internally	Hire	Cost (₹L)
Cloud	2	W6	5	2 idle staff (4 wks)	–	1.0
Data Eng	3	W3	10	3 idle staff (4 wks)	–	1.5
ML/AI	4	W4	9	–	4 permanent (from W9)	8.0
Cybersecurity	1	W6	5	–	1 contract (from W4)	3.0

Idle staff already hold level-2 skills adjacent to the gap, so training costs about ₹0.5 L each against ₹2–3 L to hire. Re-running the allocation cuts unfilled demand from 79 to 29 FTE-weeks.

4 · Project delivery risk

Project	Deadline	Staffed (now)	Risk (now)	Staffed (after plan)	Risk (after plan)
Telecom Portal Revamp	W8	100%	Low	100%	Low
Banking Cloud Migration	W10	100%	Low	100%	Low
Retail Data Platform	W12	93%	Medium	99%	Low
Insurance Claims AI	W12	45%	High	78%	High
Security Compliance Program	W12	73%	High	91%	Medium

5 · SkillSync AI agent

> **Paper vs reality:** 60 employees cover a peak demand of 56 FTE, yet 79 FTE-weeks of project work are unstaffed. The problem is skill mismatch, not headcount.

> **Critical gaps:** ML/AI (36 FTE-weeks, from W4), Data Eng (28 FTE-weeks, from W3), Cloud (10 FTE-weeks, from W6). Meanwhile QA Automation (10.4 idle), Java (9.1 idle), React (5.4 idle) have spare qualified capacity.

> **Plan:** cross-train idle adjacent-skill staff first, use contract hires for short gaps and permanent hires for long ones. Cost ₹13.5 L; unfilled demand falls from 79 to 29 FTE-weeks.

Ask: which skills are short? train or hire? which project is at risk? Ask

How it works (algorithm)

1) Integrate skill profiles (proficiency 1–5), project plans, allocations, LMS and hiring data → 2) Forecast demand per skill per week from project requirements → 3) Match supply: an employee counts only if qualified (level ≥ 3) and free → 4) Allocate scarcest skills first, using the least versatile qualified people and project priority (greedy assignment; Hungarian/min-cost flow in production) → 5) Gap = demand – staffed, per skill and week → 6) Close gaps in order of cost and lead time: (a) internal reallocation, (b) train idle adjacent-skill employees (level 2 → 3, 4 weeks), (c) contract hire (3 weeks) for short gaps, (d) permanent hire (8 weeks) for long gaps → 7) Re-run the allocation to verify the plan and the reduction in delivery risk.